

Functional Requirements

What the system does (or is designed to do) for each actor — students, teachers, and admins — organized by module. Each requirement is tagged **existing** if it is implemented in the codebase today, or **proposed** if it belongs to the redesign discussed alongside the database design and class diagrams.

Companion to [student-manager-db-design.pdf](#) and [student-manager-uml-class-diagram.pdf](#).

1. Authentication & Access Control

Covers account creation, login, and who is allowed to see or change what.

ID	Requirement	Actor	Status
FR-AUTH-01	A visitor can self-register an account with a username, email, and password.	Visitor	existing
FR-AUTH-02	A self-registered account is always assigned the STUDENT role; the client cannot request a different role.	System	existing
FR-AUTH-03	A user can log in with username and password and receive a signed JWT.	Any user	existing
FR-AUTH-04	Login fails with the same error for an unknown username and a wrong password, so a caller cannot enumerate accounts.	System	existing
FR-AUTH-05	A deactivated account cannot log in even with a correct password.	System	existing
FR-AUTH-06	Every endpoint (except auth, health, and docs) requires a valid JWT.	System	existing
FR-AUTH-07	A STUDENT can view their own student profile (/students/me) without seeing the full roster.	Student	existing
FR-AUTH-08	A STUDENT can view only their own enrollment list; requesting another student's enrollments is rejected.	Student	existing
FR-AUTH-09	TEACHER and ADMIN accounts cannot be created via self-registration; only an ADMIN can provision them, via a one-time invite code tied to a specific profile.	Admin	proposed
FR-AUTH-10	Registering with an invite code assigns the role and links the new account to the profile the code was issued for — the role is never taken from the request body.	System	proposed

2. Student Management

The academic profile record, independent of whether the student has a login account.

ID	Requirement	Actor	Status
FR-STU-01	An ADMIN can create a student record (name, matriculation number, birth date, email).	Admin	existing
FR-STU-02	Student email and matriculation number must each be unique across all students.	System	existing
FR-STU-03	An ADMIN can update or delete any student record.	Admin	existing
FR-STU-04	TEACHER and ADMIN can list the full student roster and fetch any student by ID.	Staff	existing
FR-STU-05	A student profile may exist with no linked login account (e.g. pre-registered by an admin before the student signs up).	Admin	existing
FR-STU-06	A student profile can be linked to at most one login account, via a direct reference rather than matching on email.	System	proposed

3. Course Management

The catalog of courses that can be enrolled in.

ID	Requirement	Actor	Status
FR-CRS-01	Staff (TEACHER or ADMIN) can create, update, or delete a course (code, title, description, credit hours, status).	Staff	existing
FR-CRS-02	Any authenticated user can browse the course catalog.	Any user	existing
FR-CRS-03	A course has a status of ACTIVE, INACTIVE, or ARCHIVED.	System	existing
FR-CRS-04	A course code must be unique within its academic term, so the same course can be re-offered each term.	System	proposed
FR-CRS-05	Each course belongs to exactly one academic term.	System	proposed
FR-CRS-06	Each course may have one assigned teacher; a course may be unstaffed until assigned.	Admin	proposed

4. Enrollment Management

Linking a student to a course, and tracking that relationship's lifecycle.

ID	Requirement	Actor	Status
FR-ENR-01	Staff can enroll a student in a course.	Staff	existing
FR-ENR-02	A student cannot be enrolled in the same course more than once.	System	existing
FR-ENR-03	An enrollment has a status of PENDING, CONFIRMED, or CANCELLED, staff-controlled via confirm/cancel actions.	Staff	existing
FR-ENR-04	Staff can record a grade (A, B, C, D, or F) on an enrollment, but only once it is CONFIRMED.	Staff	existing
FR-ENR-05	Cancelling an enrollment resets its grade to NOT_GRADED, so a CANCELLED enrollment can never carry a letter grade.	System	existing
FR-ENR-06	Only an ADMIN can delete an enrollment; staff can list, confirm, cancel, and grade.	Admin	existing
FR-ENR-07	A student can view their own enrollment list; staff can view any student's or any course's enrollment list.	Student / Staff	existing

5. Teacher Management

Gives the TEACHER role an actual profile and course ownership — today TEACHER is only a login role with no such record.

ID	Requirement	Actor	Status
FR-TCH-01	An ADMIN can create a teacher profile (name, email, department).	Admin	proposed
FR-TCH-02	Teacher email must be unique across all teachers.	System	proposed
FR-TCH-03	A teacher profile may be linked to at most one login account, via a direct reference.	System	proposed
FR-TCH-04	A teacher can be assigned as the owner of one or more courses.	Admin	proposed

6. Term Management

Scopes courses and enrollments to an academic period — today courses and enrollments are open-ended with no such period.

ID	Requirement	Actor	Status
FR-TRM-01	An ADMIN can define an academic term (name, start date, end date).	Admin	proposed
FR-TRM-02	A term name must be unique (e.g. "Fall 2025").	System	proposed
FR-TRM-03	Every course must be associated with exactly one term.	System	proposed

7. Account Provisioning (Registration Invites)

Replaces the fragile email-matching link with an explicit, admin-controlled claim process.

ID	Requirement	Actor	Status
FR-REG-01	Creating a student or teacher profile generates a one-time registration code tied to that specific profile and a role.	System	proposed
FR-REG-02	A registration code expires after a set period if unused.	System	proposed
FR-REG-03	A registration code can be used at most once; a used or expired code is rejected.	System	proposed
FR-REG-04	Claiming a valid code creates the login account, assigns the role from the invite, and links the account to the target profile.	Visitor	proposed

Cross-cutting business rules worth flagging

- Enrollment *status* and *grade* are independent fields, but `EnrollmentServiceImpl.cancel()` does enforce the rule in application code: cancelling an enrollment explicitly resets its grade to NOT_GRADED (so CANCELLED + a letter grade can never coexist). This is a service-layer check, not a database constraint — a direct row update could still violate it.
- Uniqueness is enforced per-entity (student email, matriculation number, course code, teacher email, term name) but never across entities — e.g. nothing stops a teacher and a student from sharing the same email today.
- All authorization is role-based (STUDENT / TEACHER / ADMIN) plus one ownership check (a student's own data); there is no per-course or per-term permission scoping.